Learning syntactic patterns for automatic hypernym discovery

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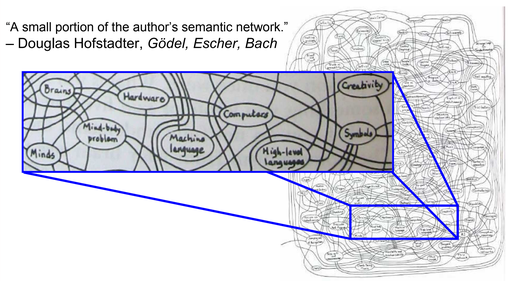
Abstract

We present a new algorithm for learning hypernym (is-a) relations from text, a key problem in machine learning for natural language understanding. This method generalizes earlier work that relied on hand-built lexico-syntactic patterns by introducing a general-purpose formalization of the pattern space based on syntactic dependency paths. We learn these paths automatically by taking hypernym/hyponym word pairs from WordNet, finding sentences containing these words in a large parsed corpus, and automatically extracting these paths. These paths are then used as features in a high-dimensional representation of noun relationships. We use a logistic regression classifier based on these features for the task of corpus-based hypernym pair identification. Our classifier is shown to outperform previous pattern-based methods for identifying hypernym pairs (using WordNet as a gold standard), and is shown to outperform those methods as well as WordNet on an independent test set.

Motivation

- It has long been a goal of AI to automatically acquire structured knowledge directly from text, e.g. in the form of a semantic network.

"A small portion of the author's semantic network."
– Douglas Hofstadter, *Gödel, Escher, Bach*



- To date, large-scale semantic networks have mostly been constructed by hand. (e.g. **WordNet**).
- We present an automatic method for semantic classification that may be used for semantic network construction; this method outperforms WordNet on an independent evaluation task.

Purpose

We aim to classify whether a noun pair (*X*, *Y*) participates in one of the following semantic relationships:

Hypernymy (ancestor)

$Y \underset{H}{>} X$ if "*X* is a kind of *Y*".

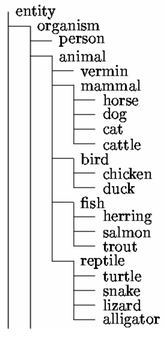
$entity \underset{H}{>} organism \underset{H}{>} person$

Coordinate Terms (taxonomic sisters)

$Y \underset{C}{\sim} X$ if *X* and *Y* possess a common hypernym, i.e. $\exists Z$ such that "*X* and *Y* are both kinds of *Z*."

$horse \underset{C}{\sim} dog \underset{C}{\sim} cat$

Once constructed, such a classifier may be used to extend semantic taxonomies such as WordNet, or create novel semantic taxonomies similar to Caraballo's hierarchy (at right).



A subset of the 'entity' branch in Caraballo's hierarchy (2001). WordNet is a hand-constructed taxonomy possessing these and other relationships for over 200,000 word senses.

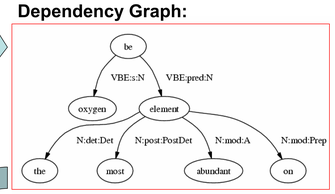
Dependency Paths as Features

For every noun pair in a large newswire corpus we use as features 69,592 of the most frequent directed paths (with redundant 'satellite' links of length 1) occurring between noun pairs in MINIPAR syntactic dependency graphs. MINIPAR is a principle-based parser (Lin, 1998) which produces a dependency graph of the form below:

Example Sentence:

"**Oxygen is the most abundant element on the moon.**"

Dependency Graph:



Dependency Paths (for "oxygen / element"):

-N:s:VBE, "be" VBE:pred:N
-N:s:VBE, "be" VBE:pred:N,(the,Det:det:N)
-N:s:VBE, "be" VBE:pred:N,(most,PostDet:post:N)
-N:s:VBE, "be" VBE:pred:N,(abundant,A:mod:N)
-N:s:VBE, "be" VBE:pred:N,(on,Prep:mod:N)

Noun Pairs as Feature Vectors

- Each noun pair ***x*** is represented as a 69,592-d vector
- Each entry x_i is the # of times feature ***i*** occurs with ***x***
- $>10^6$ vectors collected from newswire corpora comprising over six million sentences (TIPSTER 1-3 and TREC 5)
- Wikipedia used in most recent experiments

Training and Development Sets (WordNet Labels)

Training set size:	Newswire	+Wikipedia
• Hypernym:	14,387	>60,000
• Not-Hypernym:	737,924	>1,000,000

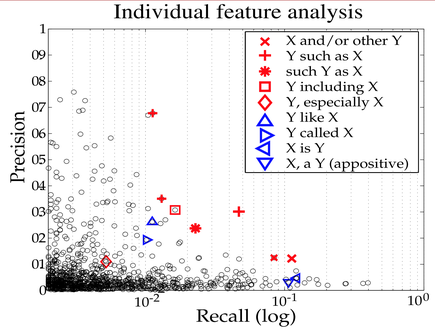
- Noun pairs labeled as "hypernym" or "not-hypernym"
- WordNet labels provide a training / development set
- All ancestors allowed as hypernyms – not just direct parents

Test Sets (Human Labels):

Test set size:	Examples	Agreement
• Hypernym:	134	82%
• Coordinate:	131	64%
• Neither:	5122	--

- Four labelers hand-labeled test set of 5,387 noun pairs
- Pairs drawn from random paragraphs in news corpus
- Each pair labeled "hypernym", "coordinate", or "neither"

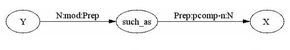
Individual feature analysis



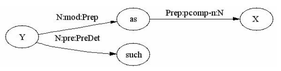
- Precision/recall for 69,592 classifiers (one per feature)
- Classifier *f* classifies noun pair ***x*** as hypernym iff $x_f > 0$
- **In red:** patterns originally proposed in (Hearst, 1992)

Rediscovering Hearst's Patterns

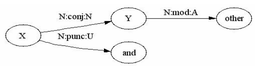
Y such as X…



Such Y as X…



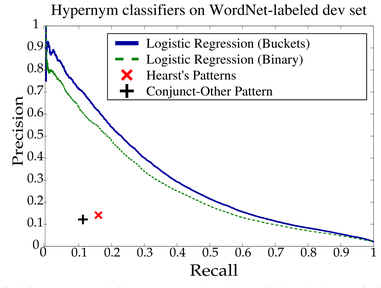
X… and other Y



Proposed in (Hearst, 1992) and used in (Caraballo, 2001), (Widdows, 2003), and others – but what about the rest of the lexico-syntactic pattern space?

A better hypernym classifier

Hypernym classifiers on WordNet-labeled dev set



- 10-fold cross validation on the WordNet-labeled data
- **Conclusion:** 70,000 features are more powerful than 6

Example: Using the "Y called X" Pattern for Hypernym Acquisition

MINIPAR path: -N:desc:V,call,call,-V:vrel:N → "<hypernym> 'called' <hyponym>"

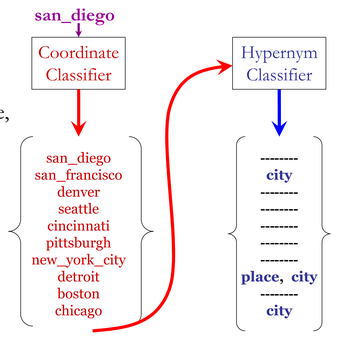
None of the following links are contained in WordNet (or the training set).

Hyponym	Hypernym	Sentence Fragment
efflorescence	condition	…and a **condition called efflorescence**…
'neal_inc	company	…The **company**, now **called O'Neal Inc.**…
hat_creek_outfit	ranch	…run a small **ranch called** the **Hat Creek Outfit**.
tardive_dyskinesia	problem	... irreversible **problem called tardive dyskinesia**…
hiv-1	aids_virus	…infected by the **AIDS virus**, **called HIV-1**.
bateau_mouche	attraction	…sightseeing **attraction called** the **Bateau Mouche**...
kibbutz_malkiyya	collective_farm	…Israeli **collective farm called Kibbutz Malkiyya**…

Hybrid Classification: Motivating Example

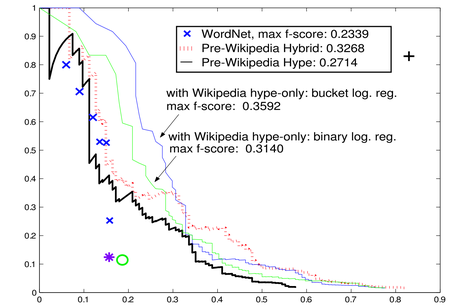
- Within-sentence hypernym data is very sparse, but distributional similarity info is plentiful
- Hybrid hypernym/coordinate classification can potentially greatly improve recall
- We define $\hat{P}\left(e_i \underset{C}{\sim} e_k\right)$ as proportional to the similarity metric used in CBC (Pantel, 2003)
- We re-estimate hypernym probabilities in the following manner:

$$\hat{P}_{new}\left(e_i \underset{H}{<} e_j\right) = \lambda_1 \hat{P}_{old}\left(e_i \underset{H}{<} e_j\right) + \lambda_2 \sum_k \hat{P}\left(e_i \underset{C}{\sim} e_k\right) \hat{P}_{old}\left(e_k \underset{H}{<} e_j\right)$$



Example: If we are uncertain of a hypernym for "San Diego", we may look to the probable hypernyms of its coordinates – and infer that "San Diego" is similarly a "city" or "place".

Hypernym Evaluation on Hand-labeled Test Set



Sample 'Additions' to WordNet

Novel Words and Links

John F. Kennedy / president
Hubei / province
Diamond Bar / city
Marlin Fitzwater / spokesman

Novel Links (Known Words)

rance / place	Reagan / leader
oybean / crop	Mark / currency
arthquake / disaster	inflation / growth
zechoslovakia / country	cat / pet

54% relative improvement over the best WordNet Classifier

Conclusion: Automatic methods can perform better than WordNet

Building a Semantic Taxonomy

Using this classifier we may now extend and construct semantic taxonomies.

We assume that the semantic taxonomy is a directed acyclic graph *G*;

We then consider the set *D* of probabilities $P\left(e_i \underset{H}{<} e_j\right)$ given by our classifier as noisy observations of the corresponding ancestry relations.

We condition the probability of our observations given a particular DAG *G*:

$$P(D \mid G) = \prod_{\substack{e_i, e_j : i \neq j, \\ e_j \in Successors(e_j)}} P\left(e_i \underset{H}{<} e_j\right) \prod_{\substack{e_i, e_j : i \neq j, \\ e_j \notin Successors(e_j)}} \left(1 - P\left(e_i \underset{H}{<} e_j\right)\right)$$

here we take the product over *all pairs* of words e_i, e_j (or synsets, in WordNet).

Our goal is to return the graph that maximizes this probability.

Algorithm: at each step we add the single link $e_i \underset{H}{\leftarrow} e_j$ that maximizes the change in probability $\delta_{e_i \underset{H}{\leftarrow} e_j}(G)$, where:

$$P(D \mid G') = P(D \mid G)\, \delta_{e_1 \underset{H}{\leftarrow} e_2}(G)$$

$$\delta_{e_i \underset{H}{\leftarrow} e_j}(G) = \prod_{\substack{e_s \in \{e_i \text{ Successors}(e_i)\} \\ e_p \in \{e_j \text{ Predecessors}(e_j)\} \\ e_c \notin Successors(e_p)}} \frac{P\left(e_s \underset{H}{<} e_p\right)}{1 - P\left(e_s \underset{H}{<} e_p\right)}$$

We continue adding links $e_i \underset{H}{\leftarrow} e_j$ so long as $\delta_{e_i \underset{H}{\leftarrow} e_j}(G) > 1$.

We have begun constructing these extended taxonomies; we plan to release the first of these for use in NLP applications in early 2005. Please let us know if you're interested in an early release!